AICE 2009 Lab 5

Rasul Abbaszada

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1 Convolution Reverb and RT60 Estimation

Convolving a signal $x[n]$ with a room impulse response (RIR) $h[n]$ produces a reverberant signal because linear convolution mathematically applies the acoustic properties of the room directly to the audio. The RIR $h[n]$ physically encodes the direct sound arrival and the precise timing and amplitude of all subsequent acoustic reflections. In the time-domain waveform, this convolution manifests as an exponentially decaying tail extending well beyond the original signal. In the spectrogram, the signal's energy is smeared horizontally across time, with different frequencies decaying at varying rates based on the absorption characteristics of the room's materials.

The RT60 estimates derived from the different measurement ranges are similar but not perfectly identical, yielding 4.02 s from the T20 fit and 4.06 s from the T30 fit. The T30 method measures a 30 dB decay and extrapolates by a factor of two, which theoretically makes it a more accurate representation of the true 60 dB decay. However, T20 measures a 20 dB drop and extrapolates by three.

2 Linear vs. Circular Convolution

To guarantee that an FFT-based multiplication produces a true linear convolution rather than a circular one, the FFT length N must satisfy the condition $N \geq L_x + L_h - 1$. In this experiment, the required minimum length to encompass the full convolution was $619,312$ samples.

If N is chosen to be too small, such as $N = \max(L_x, L_h) = 385,440$, circular convolution occurs instead. The observable time-domain problem is time-aliasing. Because the mathematically computed reverberant tail exceeds the chosen length N, the excess audio wraps around the beginning of the sequence, interfering with the start of the signal.

3 RIR Magnitude, Phase, and Group Delay

The magnitude response tells us which frequencies the room naturally amplifies or absorbs. Group delay is calculated as the negative derivative of the phase ($-\frac{d\phi}{d\omega}$) because phase mathematically represents a time shift; taking this derivative measures the exact time delay each frequency gets. To find this delay accurately, we unwrap the phase first to remove artificial 2π jumps, which would otherwise cause incorrect spikes in the derivative. The phase is then smoothed because taking a derivative amplifies high-frequency noise, and smoothing gives us a clearer overall trend. We also mask the low-magnitude frequency bins because frequencies with very little energy are mostly background noise, making their phase data unreliable.

The direct arrival time (0.000625 s) is the time it takes for the first, straight-line sound to reach the microphone. The energy centroid time (0.27 s) is the "center of mass" for the sound energy. It occurs much later because most of the room's energy arrives as delayed reflections. Finally, a large positive group delay means those specific frequencies are delayed more than others. In the time domain, this means their energy arrives later and lingers longer in the reverberant tail.

4 Deconvolution and Noise (Naive vs. Wiener)

Naive inverse deconvolution is unstable because room responses have frequencies where the energy ($H(f)$) is close to zero. Dividing by near-zero values creates massive gain spikes. When noise is added to the signal, these spikes heavily amplify the noise and ruin the output. For example, with a 20 dB input SNR, the naive method resulted in a corrupted raw output SNR of -25.15 dB.

The Wiener filter solves this problem by adding a constant α to the denominator. This stops the denominator from ever reaching zero, which limits the maximum gain and prevents the massive noise spikes seen in the naive method. Before we apply this deconvolution, we align the RIR so the recovered signal does not suffer from artificial delays or circular shifting. When checking the final quality, aligned-and-scaled SNR is often higher than raw SNR because raw SNR heavily penalizes harmless timing or volume shifts, while aligned-and-scaled SNR only measures the actual shape of the recovered wave.

In the experiment, an α_{rel} of 10^{-2} gave the best results for both noise levels (11.67 dB and 10.27 dB SNR). If α_{rel} is too small, the filter acts just like the unstable naive inverse and amplifies noise. If α_{rel} is too large, the filter reduces the noise but also removes too much of the actual signal, failing to deconvolve it properly.

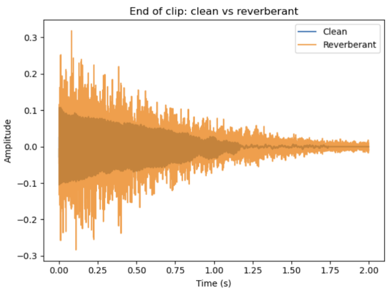

Figure 1: Clean vs. Reverb

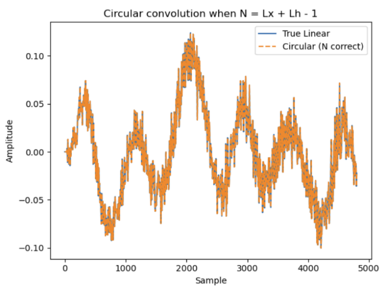

Figure 2: Circular Convolution

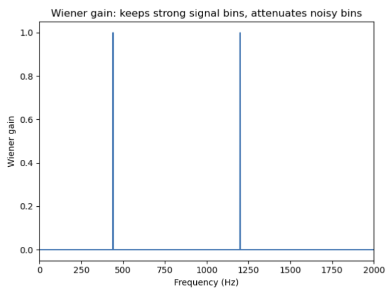

Figure 3: Wiener Gain

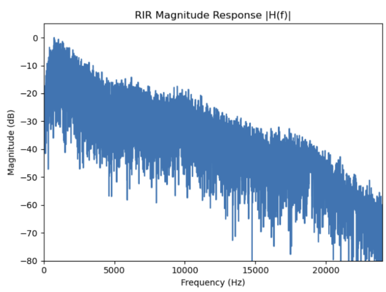

Figure 4: RIR Magnitude Response

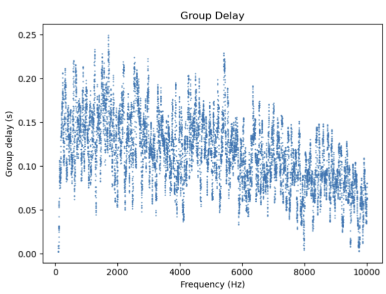

Figure 5: Smoothed Group Delay

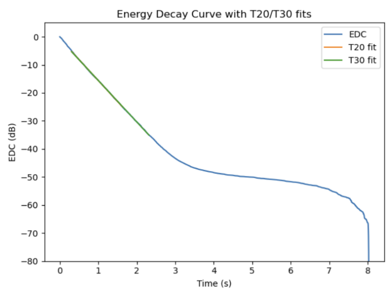

Figure 6: Energy Decay Curve & Fits